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Tep report
On

MEDICINE DISPENSER

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Abstract

With this medication dispenser, the user's card, which is provided by the administrator, stores digital points that are used to handle money. The user can replenish the points by going to the card administrator and using his identity to add new digital points to the card.

The primary objective of the dispensing machine is to supply medicine to users around-the-clock, thereby assisting the rural population who live far from hospitals or clinics in purchasing medication. Additionally, it can be installed similarly to an ATM, making it simple for people to use even if they are blind or visually impaired.

This paper presents the design and implementation of medicine dispenser that utilises Arduino UNO, LCD Display, Push buttons, Servo motor, RFID. The Arduino UNO servers as the central microcontroller, connecting the various components.

Radio-frequency identification (RFID) is a wireless technology that transfers data from an electronic tag attached to an object through a reader using radio waves. RFID presents an approach to identify and manage tools and equipment.

Keywords: RFID (radio frequency identification), Arduino UNO, Motor driver set, and the Internet of Things.

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MODULE 1

INTRODUCTION

The care card, which is used to store user data, is used by the pharmaceutical distribution system. Here, we're employing an RFID tag to identify the user.

This tag tracks and identifies the user's records, including login information, medication intake, and quantity, all of which are stored for future reference using radio frequency identification devices. The read/write device, the tag itself, and a host system application for data processing, transmission, and collecting comprise the RFID tag system.

The Arduino UNO is the main system controller used by the medication dispenser. It allows all of the modules to be attached and perform their respective functions. Since the machine needs more pins, we chose the Arduino UNO as the controller. The four essential modules that are attached to this controller are the LCD display, servo motor and keypad.

In underdeveloped nations, elderly care is a major concern. The elderly must be managed and cared by family members. It is challenging for family members to care for the elderly 24/7 in the modern world. In today's society, the majority of families are nuclear.

The elderly have a natural yearning for independence and would want to stay that way, but their offspring are concerned. Even with the best of intentions, elderly people occasionally forget to take their medications on schedule. One such method to assist them in taking their medications effectively is the automatic medication dispenser.

People are under increasing pressure to choose a medication-administration gadget as the expense of in-home healthcare continues to grow. The automated medication dispenser accomplishes its goal.

An innovative piece of technology created to improve and expedite the distribution of medications is the automatic medicine dispensing machine. It reduces human error in the precise dispensing of prescription medications by utilizing robotics and smart technology. To guarantee authorized access, these devices frequently have secure user authentication installed. Patients can easily access their medications thanks to the user-friendly design.

Through the reduction of waiting periods, this kind of innovation enhances healthcare efficiency. Additionally, the automated dispensing procedure helps to provide a more sterile and hygienic atmosphere for handling medications. All things considered, these devices serve as evidence of how healthcare and technology may work together to improve patient outcome.

MODULE 2

BLOCK DIAGRAM

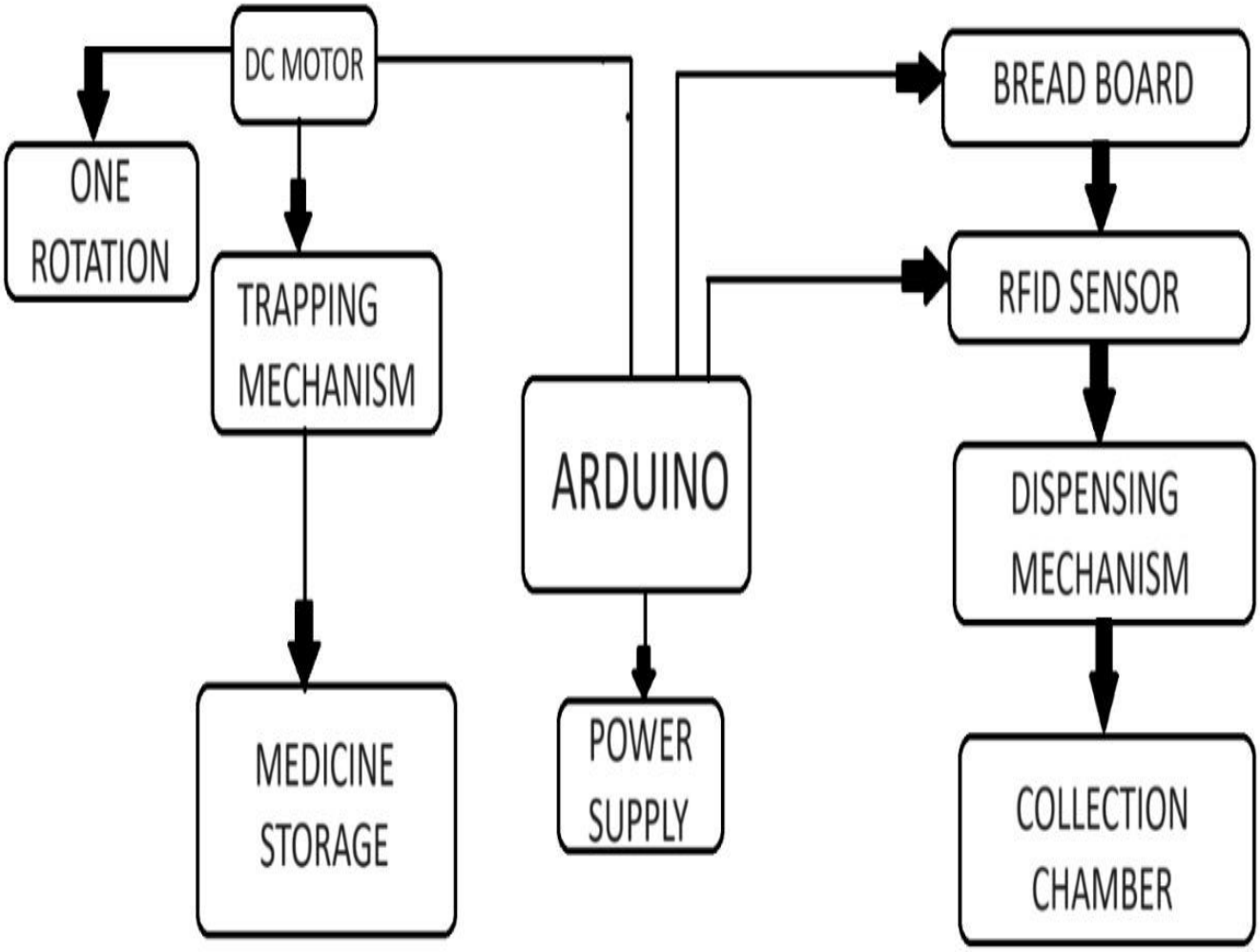

Figure 1: Block Diagram

MODULE 3

LIST OF THE COMPONENTS

1. Arduino UNO
2. DC Motor
3. Push Buttons
4. LCD Display
5. RC522 RFID
6. Bread Board
7. RFID Card
8. Header Pins
9. Relay Module
10. 9 V Battery

3.1 Arduino UNO:

The Arduino Uno is an open-source microcontroller board that was created by Arduino.cc and first made available in 2010. It is based on the Microchip ATmega328P microprocessor (MCU).

Various expansion boards (shields) and other circuits can be interfaced to the sets of digital and analog input/output (I/O) pins on the microcontroller board.

The board may be programmed using the Arduino IDE (Integrated Development Environment) and a type B USB cable. It includes six analog and fourteen digital I/O pins, six of which can be used for PWM output. It can be powered by a rectangular 9-volt battery or a barrel connector that takes voltages ranging from 7 to 20 volts.

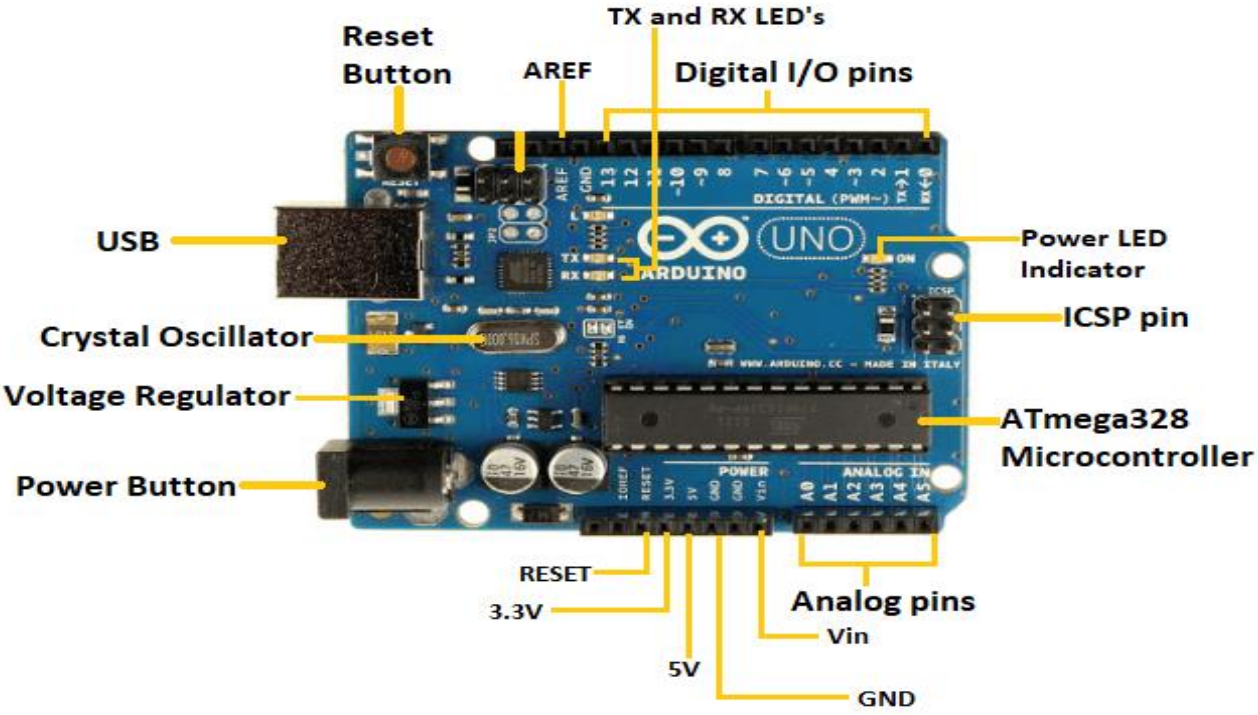

Figure 2: Arduino uno

Atmega328 Microcontroller:

The ATmega328 microcontroller belongs to the ATmel family and is a single chip microcontroller. Its processor code is eight bits long. It includes registers, I/O lines, SPI serial ports, memory (SRAM, EEPROM, and Flash), oscillator, timer, and external and internal interrupts.

ICSP pin:

The Arduino board's firmware can be programmed by the user via this pin, which stands for In-Circuit Serial Programming.

Power LED Indicator:

When an LED is in the ON state, it indicates that power is on. The LED won't turn on when the power is off.

Digital I/O pins:

These can be set to HIGH or LOW. Digital pins have numbers ranging from D0 to D13.

TX and RX LEDs:

The lighting of these LEDs indicates a successful data transfer.

AREF:

An external power supply is utilized to deliver a reference voltage to the Arduino UNO board through the Analog Reference (AREF) port.

Reset button:

This button is added to the connection for that purpose.

USB:

This enables computer connectivity for the board. It is necessary in order to program the Arduino UNO board.

Crystal Oscillator:

The Arduino UNO is a powerful board because of the 16MHz frequency of the crystal oscillator.

Voltage Regulator:

This device raises the input voltage to five volts.

Ground pins, or GND:

The ground pin functions as a zero-voltage pin.

Vin is the voltage input.

Analog Pins: Analog pins have numbers ranging from A0 to A5. Reading the analog sensor utilized in the connection is the purpose of the analog pins. Also, it is capable of functioning as GPIO (General Purpose Input Output) pins.

3.2 DC Motor :

A Direct Current (DC) motor's efficiency, controllability, and versatility make it a vital electromechanical tool used in a wide range of industries. It works on the basis of the idea that magnetic fields can interact to transform electrical energy into mechanical energy.

A DC motor's stator, rotor, commutator, brushes, and windings are its essential parts. The field windings in the stator, which is the stationary portion of the motor, generate a constant magnetic field when an electric current passes through them. Conversely, the rotating component inside the stator's magnetic field is called the rotor. The rotor rotates as a result of the electromagnetic force produced when current flows through the rotor windings.

DC motors come in a variety of forms, the most popular of which are brushed and brushless DC motors. In brushed DC motors, the rotor rotates continuously because the commutator and brushes reverse the direction of current flowing through the rotor windings. But because of brush wear, these motors need to be maintained on a regular basis. In contrast, brushless DC motors do not require brushes because they use electronic controllers to switch the current in the stator windings. They provide increased performance, lower maintenance costs, and increased efficiency.

DC motors are ideal for a variety of applications because they offer superior control over both speed and torque. They are employed in a variety of sectors, including electronics, robotics, automotive, and manufacturing.

Figure 3 : DC Motor

3.3 LCD Display:

- An electrical device that shows data and messages is called an LCD 16x2. Its 16 columns and 2 rows, as the name suggests, enable it to display a total of 32 characters (16x2=32). Every character is made up of 40 5x8 pixel dots. Consequently, 32x40 pixels, or 1280 pixels if not, can be used to calculate the total number of pixels in this LCD.

- Rather than electrons diffracting at a glass panel, a backlight arranged in a rectangular network illuminates each pixel in a liquid crystal display.

- Every pixel has a blue, red, and green sub-pixel that can be turned ON or off. When all of these sub-pixels are turned on, it will appear white, and when all of them are turned off, it will appear black.

- By changing the intensity of each light, numerous colour combinations can be produced. This article provides an overview of the LCD 16X2 and how it works with apps.

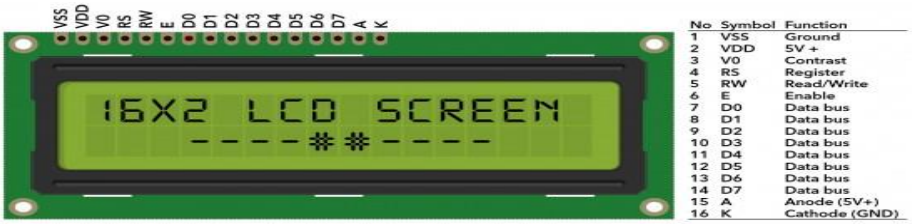

Figure 4 :LCD Screen

3.4 Push Buttons

Figure 5 : push buttons

Electrical actuators known as push button switches, or pushbutton switches, are connected to electrical circuits and function as electrical actuators by closing or opening those circuits when pressed. They can operate a large variety of electronic devices.

These switches come in the shape of keys or buttons. They may be transient or maintained. The push button is the most common type of momentary switch. Push-to-break is a term used occasionally to describe a normally closed push-button switch push-to-make is a term frequently used to describe a normally open push button switch.

3.5 RC522 RFID:

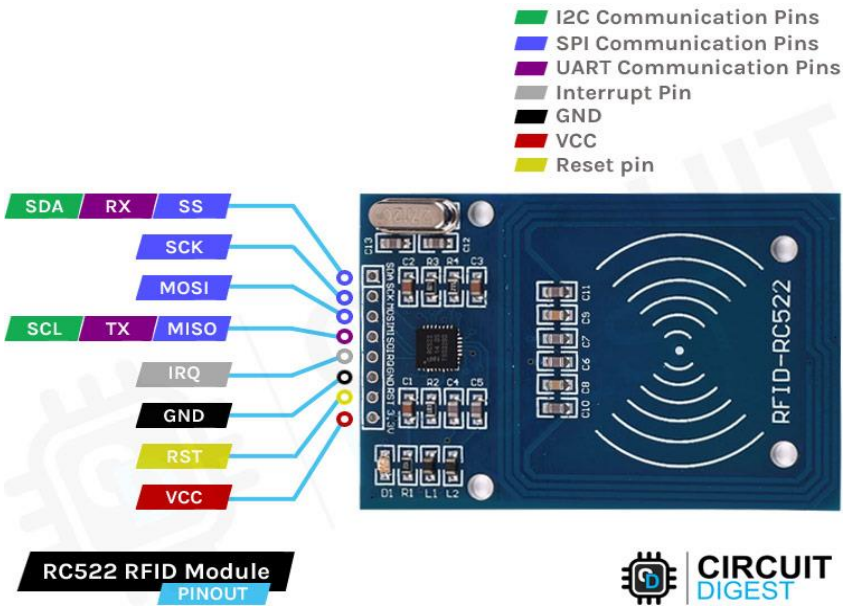

Figure 6 : RC522 RFID

Radio-frequency identification (RFID) automatically detects and tracks tags attached to objects using electromagnetic fields. A small radio transponder, a radio receiver, and a transmitter make up an RFID system. The tag returns digital data, typically an inventory number, to the RFID reader when it receives an electromagnetic interrogation pulse from a nearby device.

VCC:

VCC supplies power for the module. This can be anywhere from 2.5 to 3.3 volts. It can be connected to 3.3V output from the Arduino.

RST:

RST is an input for Reset and power-down. When this pin goes low, hard power-down is enabled. This turns off all internal current sinks including the oscillator and the input pins are disconnected from the outside world. On the rising edge, the module is reset.

GND:

GND is the Ground Pin and needs to be connected to the GND pin on the Arduino.

IRQ:

IRQ is an interrupt pin that can alert the microcontroller when an RFID tag comes into its vicinity.

MISO/SCL/Tx:

MISO / SCL / Tx pin acts as Master-In-Slave-Out when SPI interface is enabled, acts as serial clock when I2C interface is enabled and acts as serial data output when UART interface is enabled.

MOSI:

The RC522 module receives SPI input from MOSI (Master Out Slave In).

SCK:

The Arduino, which is the SPI bus master, provides clock pulses that the SCK (Serial Clock) accepts.

SS/SDA/Rx:

If the SPI interface is enabled, the SS, SDA, and Rx pins function as signal input; if the I2C interface is enabled, they function as serial data input; and if the UART interface is enabled, they function as serial data input. In order to make it easy to identify the other pins, this pin is typically marked by enclosing it in a square.

3.5 Breadboard:

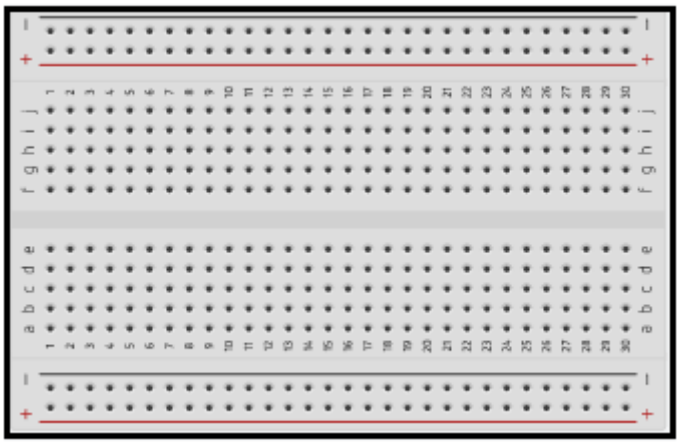

Figure 7 :bread board

- A white rectangular board with tiny embedded holes for electronic component insertion is called a breadboard. In electronics projects, it is frequently used. Another way to describe a breadboard is as a prototype that serves as the foundation for building electronics.
- The words bread and board are the origin of the word breadboard. When the bread pieces were first sliced, the term "breadboard" was used. But because it was used in electronics in the 1970s, it was also referred to as a breadboard.
- Therefore, the word "breadboard" only applies to these boards, which enable a rapid electrical connection.
- Another name for a breadboard is a solderless board. It indicates that soldering is not necessary for the component to fit into the board. As a result, the breadboard is reusable.
- By connecting the component's end terminal to the board, we can quickly and easily fit it. Thus, a breadboard is frequently referred to as a plugboard.

3.6 RFID Card:

Figure 8 : RFID card

- Radio-frequency identification is known as RFID. RFID cards are used in situations where access control is necessary or where personnel tracking or identification is crucial.
- Cards nowadays use a range of RFID frequency bands, such as 13.56 MHz high frequency smart card, 125 kHz low frequency proximity, and 860-960 MHz ultra-high frequency (UHF).
- Many times, smart cards and proximity cards are just called "RFID cards." The application determines the type of RFID frequency band to be used, taking into account the required data transfer speed, read range, and security level.
- The standard proximity card format for employee badges, door and gate access control, and other applications is 125 kHz (LF).
- Higher security format used for employee badges and credit cards for both logical and physical access control is 13.56 MHz (HF). UHF: 860–960 MHz

3.7 Header Pins

3.7.1 Male:

Figure 9: male

Male header pins are widely used electrical components that are used to interface and connect different types of electronic devices and components. Male header pins are made up of a row of stiff, projecting pins attached to metal wires. Usually composed of brass or phosphor bronze, these pins may also be plated in gold or tin for increased conductivity and resistance to corrosion.

The purpose of male header pins is to fit into female connectors or receptacles. They establish an electrical connection when they are inserted, enabling the transfer of power, signals, or data between various components.

Male header pins can be polarized, which means they have a particular orientation to guarantee correct alignment when joining parts. This aids in avoiding inadvertent reverse connections

3.7.2 Female:

Figure 10 : female

- Female header pins, sometimes referred to as female holders or socket connectors, are essential parts of electronic devices and circuits that are used to connect and interface. A row of sockets or receptacles arranged in a plastic housing makes up a female header pin. The purpose of these sockets is to receive and make contact with the male pins of various components, including boards, sensors, and microcontrollers.

3.8 Resistor:

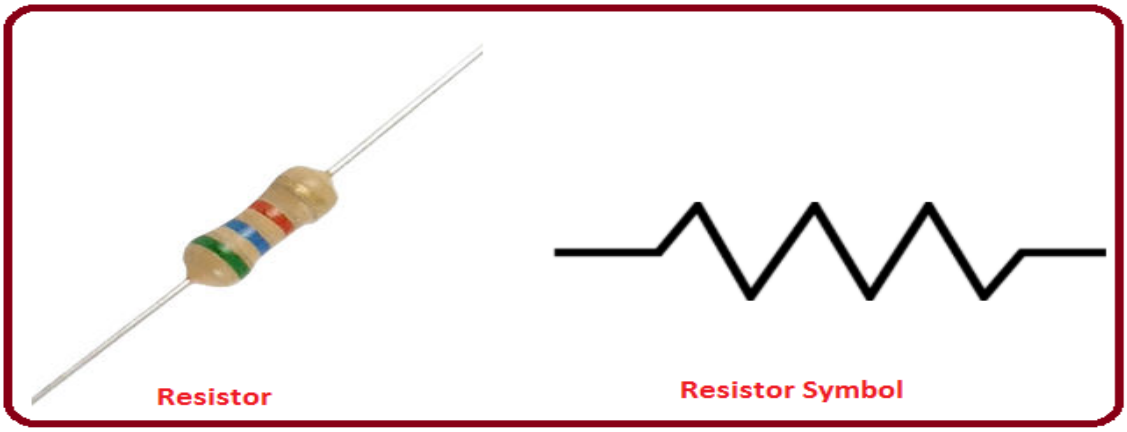

Figure 11 : resistor

- The resistor is a passive electrical component that creates resistance in the flow of electric current. In almost all electrical networks and electronic circuits they can be found.
- The resistance is measured in ohms (Ω). An ohm is the resistance that occurs when a current of one ampere (A) passes through a resistor with a one volt (V) drop across its terminals. The current is proportional to the voltage across the terminal ends.
- This ratio is represented by Ohm's law: R=VI.
- Resistors are used for many purposes. A few examples include limiting electric current, voltage division, heat generation, matching and loading circuits, gain control, and setting time constants. They are commercially available with resistance values over a range of more than nine orders of magnitude. They can be used as electric brakes to dissipate kinetic energy from trains or be smaller than a square millimetre for electronics.

3.9 Relay Module

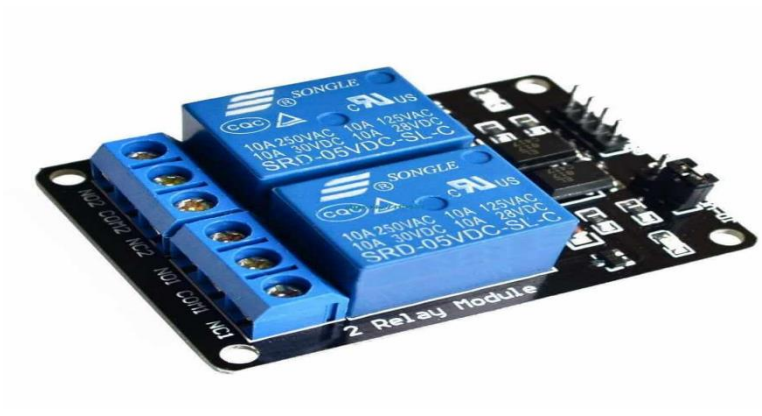

Figure 12 : Relay Module

Relay modules are electromechanical devices that use low-power control signals to operate high-power electrical devices. It functions as a switch that is signaled externally. The relay module is made up of various essential parts, including an armature, a coil, a spring, and one or more sets of contacts. It works by using electromagnetic forces to open or close its contacts in response to an applied voltage or current.

Relay modules' ability to separate the controlling circuit from the controlled circuit is their main benefit. This is essential in situations where low-power microcontrollers, Arduino boards, Raspberry Pi, or other comparable control systems must be used to operate high-power or high-voltage devices.

These sensitive control systems can safely run and regulate heavy loads, such as heaters, lights, motors, and other commercial or residential appliances, by utilizing relays.

Different configurations of relay modules are available, including double pole single throw (DPST), double pole double throw (DPDT), single pole single throw (SPST), and single pole double throw (SPDT). The configuration option selected will rely on the particular needs of the application.

Relay modules can also have varying ratings according to their voltage and current capacities, which guarantees that they will work with the intended load. Additionally, some modules might have built-in protection mechanisms to prevent reverse current or voltage spikes from harming the controlling circuitry, or opto-isolation for better electrical isolation.

Relays do have certain limitations, though, like mechanical wear from moving parts over time that could shorten their lifespan. Furthermore, in comparison to solid-state alternatives like transistors or MOSFETs, their switching speed is slower.

In conclusion, relay modules are crucial parts of electrical and electronic systems because they allow high-power devices to be safely and carefully switched by low-power control signals, which makes a variety of applications run more smoothly.

3.10 9V Battery

Figure 13: Battery

A power source that is portable and often used, a 9-volt battery produces a voltage of roughly 9 volts. The adequate energy capacity and compact size of this device make it a popular choice for a variety of electrical gadgets. This battery, which is usually rectangular in shape, frequently has six tiny cells arranged in series inside of its casing.

The 9-volt battery has several noteworthy uses, including smoke detectors, where its durability and dependability are essential to the ongoing operation of this safety device. It's also used in a variety of electrical devices, including guitar pedals, remote controllers, and tiny electronic toys.

Zinc and manganese dioxide are the two main chemical components of a typical 9-volt battery, which facilitates a reliable and effective electrochemical reaction that generates electrical energy. The battery's comparatively extended shelf life and consistent power production while in operation are made possible by this architecture.

Although 9-volt batteries are widely used, they do have several drawbacks. Compared to larger batteries, their energy capacity is moderate, therefore high-drain devices that require a lot of power might not be a good fit for them. Although they exist, rechargeable choices are less frequent because it is difficult to design rechargeable batteries with this voltage.

Because the positive and negative terminals of 9-volt batteries are near to one another and can short circuit if they come into contact with metal items, users should handle these batteries carefully in order to ensure their safety. It's also critical to dispose of these batteries properly because they can contain elements that, if not handled properly, could harm the environment.

In summary, the 9-volt battery is essential for powering a wide range of commonplace gadgets because it strikes a compromise between its small size and respectable energy production. Its position in the field of portable electronics has been cemented by its dependability and versatility.

MODULE 4:

WORKING PRINCIPLE

4.1 FLOW CHART

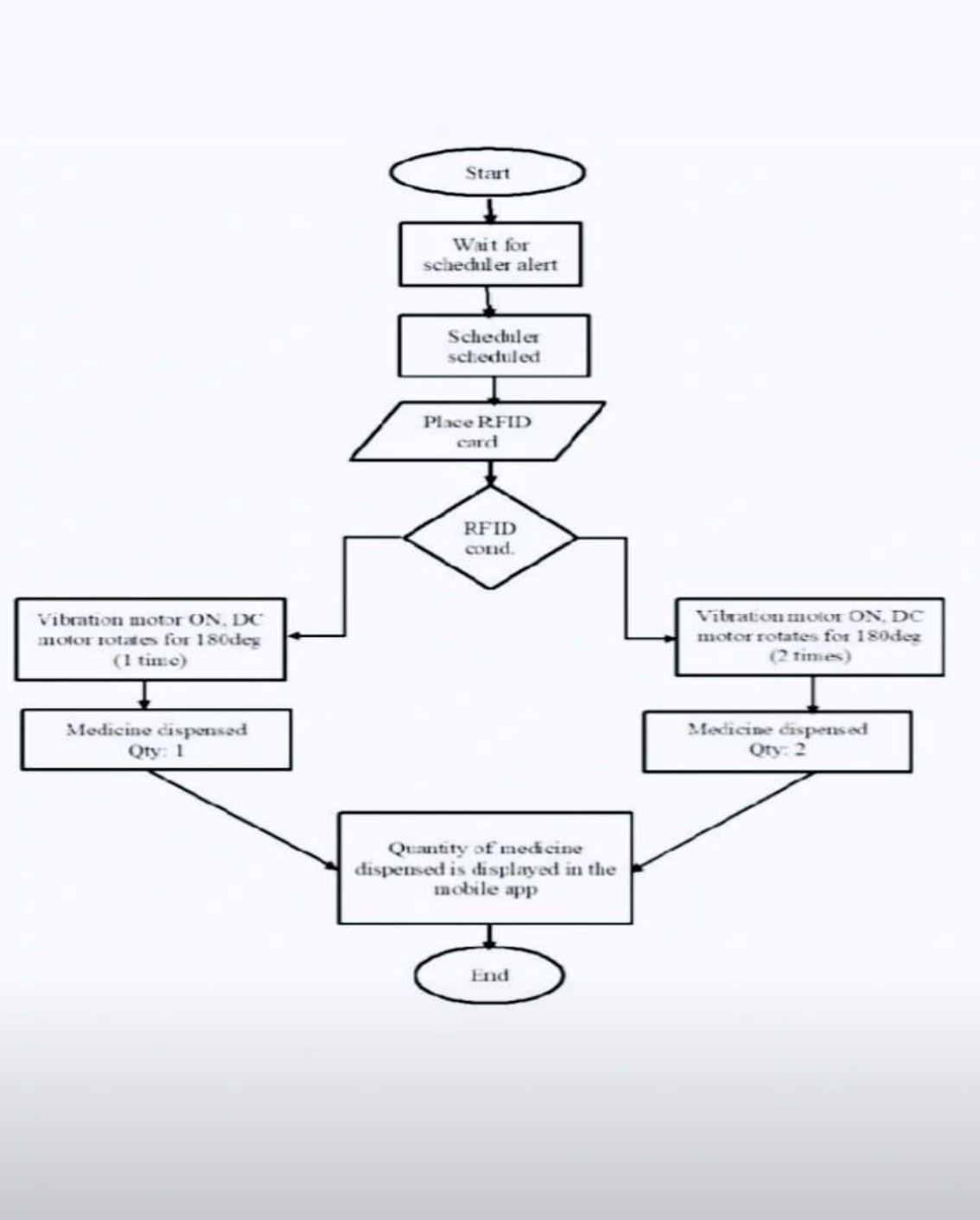

Figure 14 : Flow chart

4.2 CONNECTION DIAGRAM:

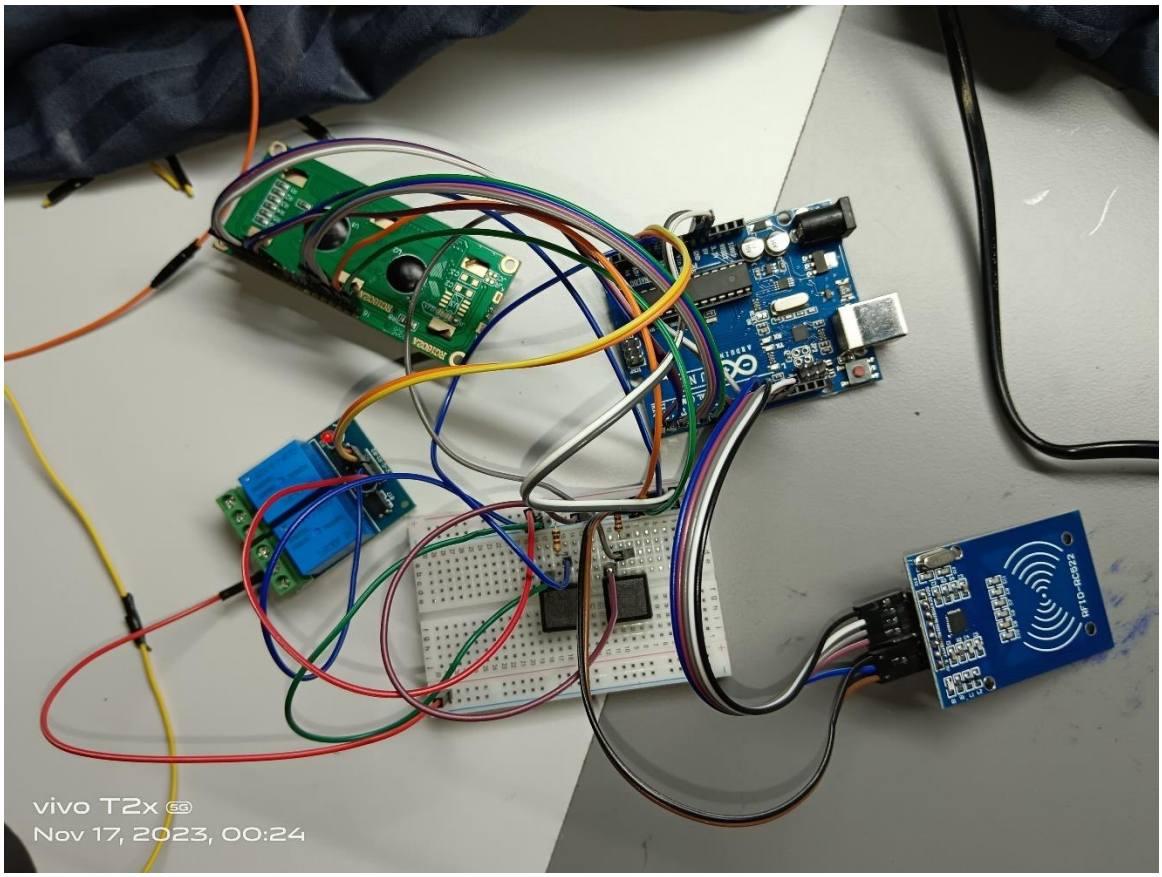

Figure 15 : Connection Diagram

4.3 CODE :

```
#include <SPI.h>

#include <MFRC522.h>

#define SS_PIN 10

#define RST_PIN 9

MFRC522 mfrc522(SS_PIN, RST_PIN);  // Create MFRC522 instance.

#include <LiquidCrystal.h>

LiquidCrystal lcd (8, 7, 5, 4, 3, 2);

int button1Pin = A0; // Button for compartment 1

int button2Pin = A1; // Button for compartment 2

#define compartment1 A2

#define compartment2 A3

 void setup ()

{

  Serial.begin(9600); // Initiate a serial communication

  SPI.begin();     // Initiate SPI bus

  mfrc522.PCD_Init ();  // Initiate MFRC522



  Serial.println("Put your card to the reader...");

  Serial.println();

  pin Mode(compartment1, OUTPUT);

  pin Mode(compartment2, OUTPUT);

  digitalWrite(compartment1, LOW);

  digitalWrite(compartment2, LOW);

    content.concat(String(mfrc522.uid.uidByte[i] < 0x10 ? " 0" : " "));

    content.concat(String(mfrc522.uid.uidByte[i], HEX));

  }

  Serial.println();

  Serial.print("Message : ");

  content.toUpperCase();

  if (content.substring(1) == "14 DC 1A 04") //change here the UID of the card/cards that you want to give access

  {

    Serial.println("Authorized access");

    Serial.println();

    lcd.clear();

    lcd.setCursor(0, 0);

    lcd.print("Authorized access");

    lcd.setCursor(0, 1);

    lcd.print("Select compartment");

    while (true) {

    if (digitalRead(button1Pin) == LOW) {

      openCompartment1();

      while (false);

    }

    else if (digitalRead(button2Pin) == LOW) {

      openCompartment2();
```

```
  // Initialize button pins
  pinMode(button1Pin, INPUT);
  pinMode(button2Pin, INPUT);
  // Initialize LCD
  analogWrite(6,130);
  lcd.begin(16, 2);
  lcd.setCursor(0, 0);
  lcd.print("Vending Machine");
  delay(2000);
}
void loop()
{
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("Place card...");
  delay(1000);
  lcd.clear();
  // Look for new cards
  if ( ! mfrc522.PICC_IsNewCardPresent())
  {
    return;
  }
// Select one of the cards
  if ( ! mfrc522.PICC_ReadCardSerial())
  {
    return;
  }
//Show UID on serial monitor
```

```
      while (false);
    }
  }
  }
   else   {
    Serial.println(" Access denied");
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("Authorized denied");
}
}
 void openCompartment1() {
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("Opening 1...");
  digitalWrite(compartment1, HIGH);
  delay(3000);  // Adjust the delay as needed
  digitalWrite(compartment1, LOW);
  lcd.setCursor(0, 1);
  lcd.print("Thank you!");
  delay(2000);
  lcd.clear();
}
 void openCompartment2() {
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("Opening 2...");
  digitalWrite(compartment2, HIGH);
```

```
  Serial.print("UID tag :");

  String content= "";

  byte letter;

  for (byte i = 0; i < mfrc522.uid.size; i++)

{

    Serial.print(mfrc522.uid.uidByte[i] < 0x10 ? " 0" : " ");

    Serial.print(mfrc522.uid.uidByte[i], HEX);
```

```
  delay(3000);  // Adjust the delay as needed

  digitalWrite(compartment2, LOW);

  lcd.setCursor(0, 1);

  lcd.print("Thank you!");

  delay(2000);

  lcd.clear();

}
```

MODULE 5

MECHANICAL DESIGN

5.1 3-D MODEL

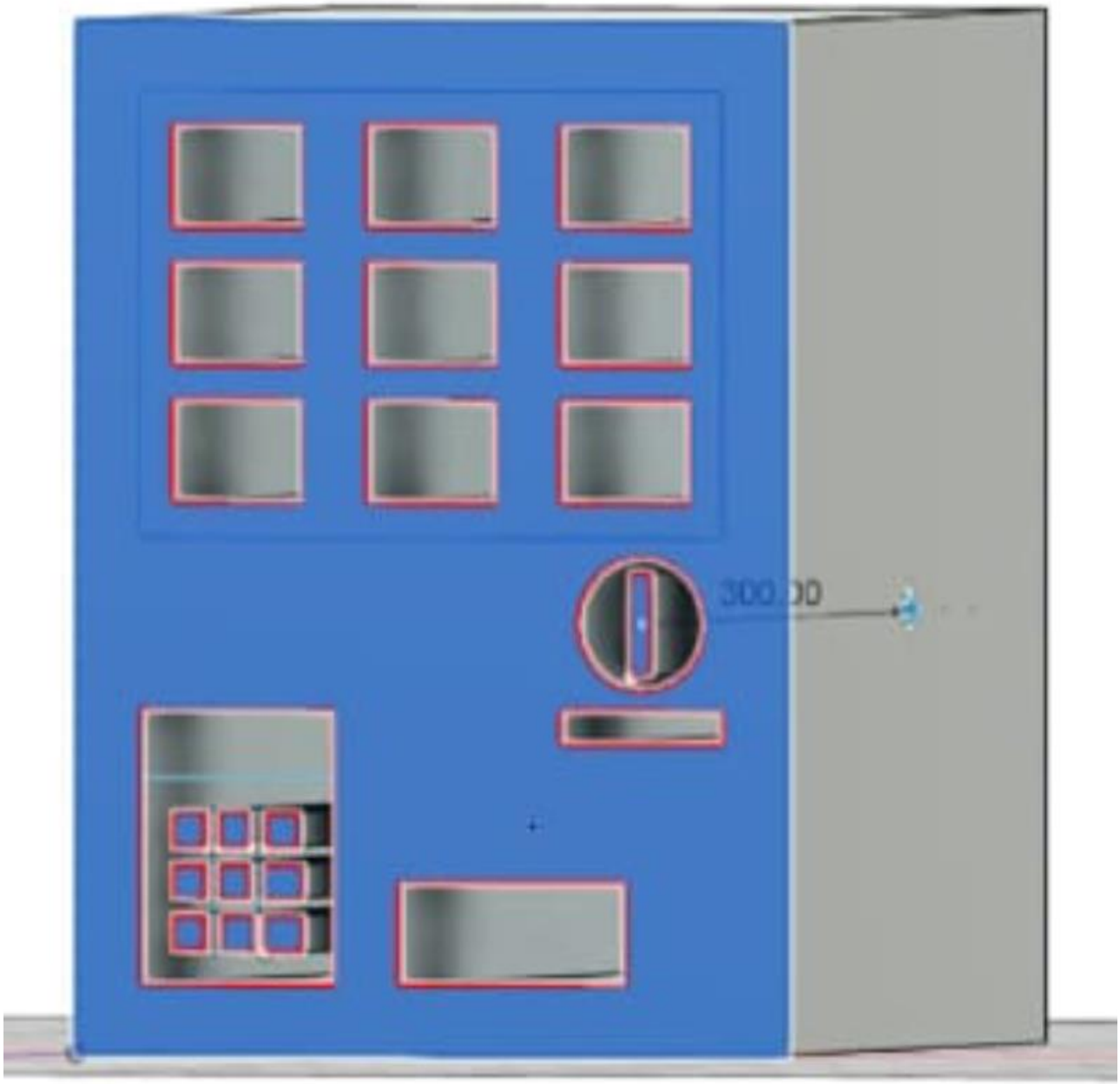

Figure 15: 3D model

EXPLANATION

A medicine dispenser is a specialized tool that helps patients properly manage their treatment regimens by organizing and dispensing medications while guaranteeing dosage adherence. These gadgets are available in a range of configurations to meet diverse requirements and complexity levels.

More advanced medicine dispensers are automatic or electronic. These devices are equipped with timers or alarms, allowing users to program specific times for medication doses. At the scheduled time, the dispenser releases the appropriate pills or dosage. Some automatic dispensers also offer features like locking mechanisms or tamper-proof compartments to prevent unauthorized access or accidental overdosing. These are particularly useful for individuals with complex medication regimens or those who require precise dosing schedules.

Smart medicine dispensers, which incorporate technology to improve medication management, are the result of recent advancements. These gadgets are frequently linked to internet portals or smartphone apps. When it's time for users to take their medication, smart dispensers notify them by sending alerts and reminders to their phones. Additionally, they can produce reports on medication adherence, giving consumers and medical professionals useful information. Furthermore, some smart dispensers have features that let family members or caregivers keep an eye on a user's medication schedule from a distance, offering comfort and assistance.

All things considered, medicine dispensers are essential for increasing medication adherence, decreasing mistakes, and streamlining the administration of numerous medications. They provide a range of services, from simple organization to cutting-edge technological solutions, with the ultimate goal of enhancing health outcomes by ensuring people take their prescription drugs as directed**.**

MODULE 6 :

APPLICATION OF MEDICINE DISPENSER:

1. Patient Adherence: Promotes better adherence to treatment plans by making sure patients take their prescription drugs on time.

2. Chronic Disease Management: Lowers the likelihood of missed doses and automates medication schedules to support people with chronic conditions.

3. Elderly Care: Offers a dependable medication management system to enable elderly people to live independently.

4. Reduced Human Error: Lowers the possibility of medication administration errors by people, enhancing patient safety.

5.Remote Monitoring: Improves patient care by enabling medical professionals or family members to remotely check on medication adherence.

MODULE 7 :

CONCLUSION :

A medicine dispenser provides a methodical way to manage prescription schedules, which is a revolutionary solution in the healthcare industry. It is a very useful tool, especially for people who have complicated medication schedules or who need help following through on prescribed treatments. This cutting-edge tool improves patient adherence, simplifies medication administration, and advances health outcomes.

The medicine dispenser is essential to its operation because it guarantees accurate dosage administration, which greatly lowers the possibility of medication errors. Medication errors, overdoses, and missed doses are reduced by compartmentalizing medications and dispensing them at predetermined times. This feature is particularly helpful for the elderly, people with long-term conditions, or people who have cognitive impairments because it provides a structured method for them to take their medications on their own.